

FALCON's Response to Glaukous For ARM sorter with Secondary Sorting 2K

February 3, 2024



www.falconautotech.com

Kind Attention –

Mr. Tanzil

M/s Glaukous

Offer Ref: F24-00122 ; Date: 03-02-2024

Subject – Techno -Commercial Offer of Arm Sorter with Secondary Sortation System 2K PPH

Dear Team,

We are pleased to submit our Techno-commercial Offer in response to your RFP

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies.

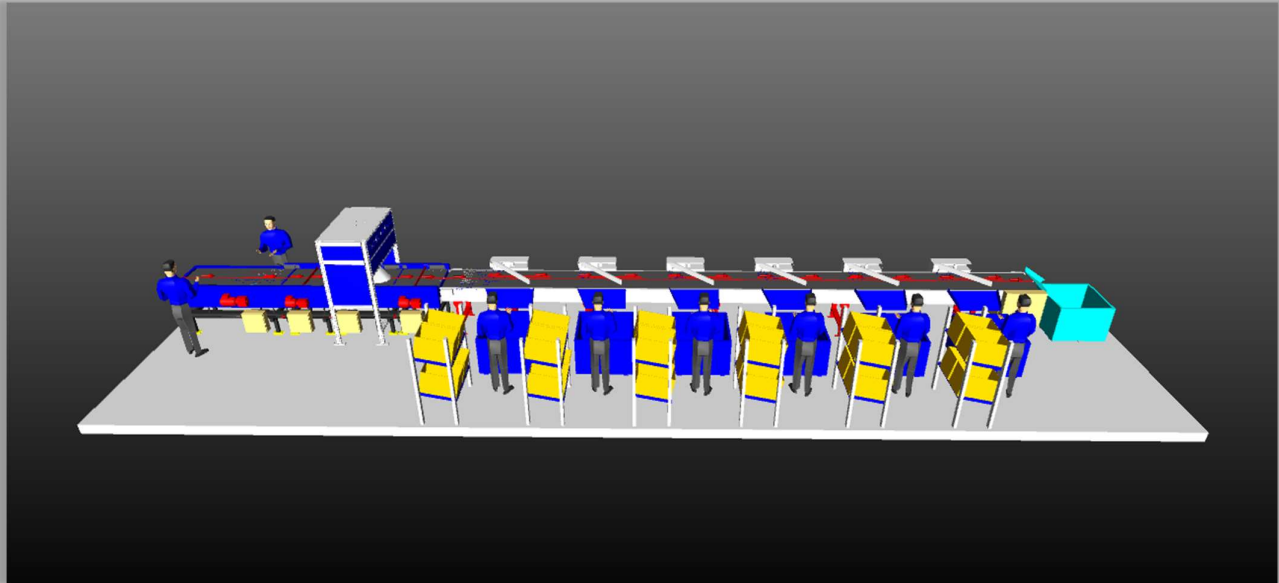
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Best Regards,

Sandeep Bansal

Chief Business Officer

Response to RFP for Arm sorter With Secondary Sortation System 2K PPH Throughput



Proposal Reference – F24-00122

**Falcon Autotech Private Limited
Plot No. 87, Sector Ecotech-1,
Extention-1, Greater Noida, Uttar
Pradesh 201308.**

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1. Glossary

S. No.	Term	Description
1	RFP	Request For Proposal
2	PPH	Parcels Per Hour
3	IT	Information Technology
4	BOQ	Bill Of Quantity
5	I/O	Input/ Output
6	PDP	Power Distribution Panel
7	PC	Personal Computer
8	UPS	Uninterrupted Power Supply

2. Executive Summary

Falcon is pleased to confirm its great interest in responding to this GLAUKOUS RFQ of Arm Sorter. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening, understanding your needs, and ensuring this project's success.

Following the same objective for the Arm Sorter system along with secondary sortation system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.

Our solution is based on the following key characteristics:

- **One Arm based sorter for shipments, employing our Electric ARM technology, have been designed for a throughput of 2,000 packages per hour on one highway line.**
- **The Sorter is equipped with its own infeed conveyor system.**
- **In the Sorter, there are 6 Collection Chute and One Sortfail chute.**
- **Further four Bins are integrated with each chute for secondary sorting.**
- **The system layout has been meticulously planned to facilitate smooth operational flow, ensuring efficient movement of personnel.**

A tailor-made and simple layout designed explicitly for GLAUKOUS. The proposed layout is the result of the technical requirements and our discussions with the relevant stakeholders during our meeting.

1. Falcon's reliable Shipment sortation systems

These systems are globally being used by most innovative brands such as Amazon, Delhivery, Asendia, Fastway and Glaukous many more. The main and critical components of the FALCON Autotech sorter building blocks, like Conveyors, belts, bearings, Communication platforms, PLCs etc., are sourced from some of the best suppliers in the world, such as SEW, Omron, Cognex, Vahle and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

2. Commitment to quality systems

Demonstrating Falcon's clear commitment to the GLAUKOUS satisfaction, the shipment sortation system, parts and services will be under warranty for 12 months from installation go-live.

3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0

Operational since 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces

Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS

Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5 Product Lines

15+ Countries with Live Installations

1800+ Total Installed Systems Globally

1Million+ Sorts Per Hour

600+ Employees

Long-standing Relationships with industry leaders

News & Articles

FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.

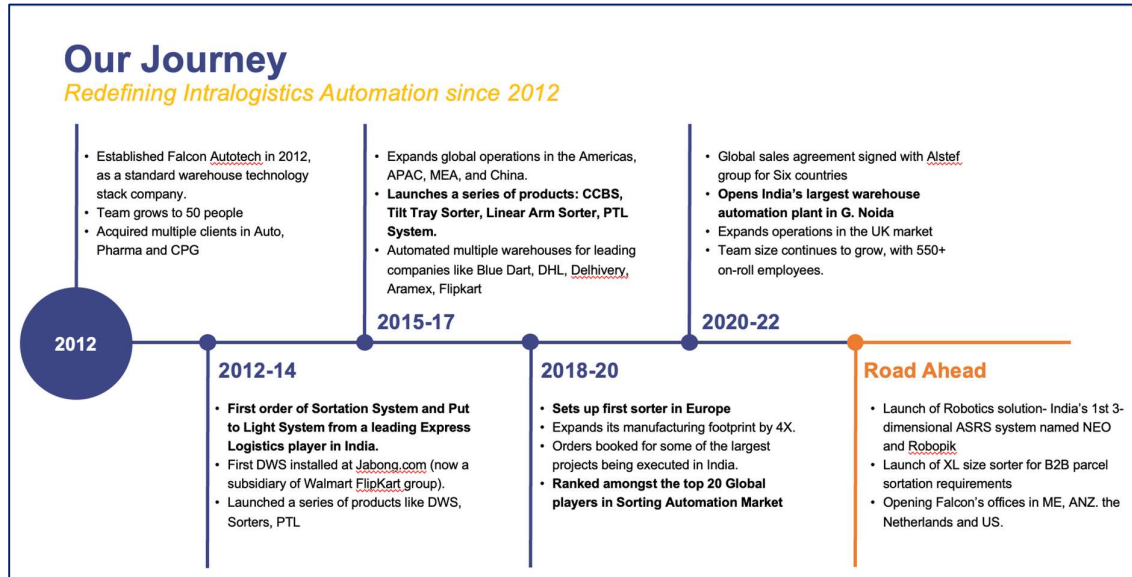
FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



Product and Solutions

With 100% focused on Parcels, Eaches, Totes, Bags, Cartons



SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- PopUp Sorter
- Sweep Sorter
- Pusher Sorter



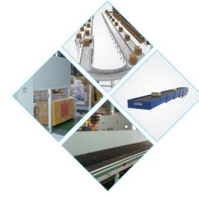
PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
 - R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
 - Mini (600MM)
 - Jumbo (1200MM)



CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



ROBOTICS

- NEO
- Robopick

Powered by

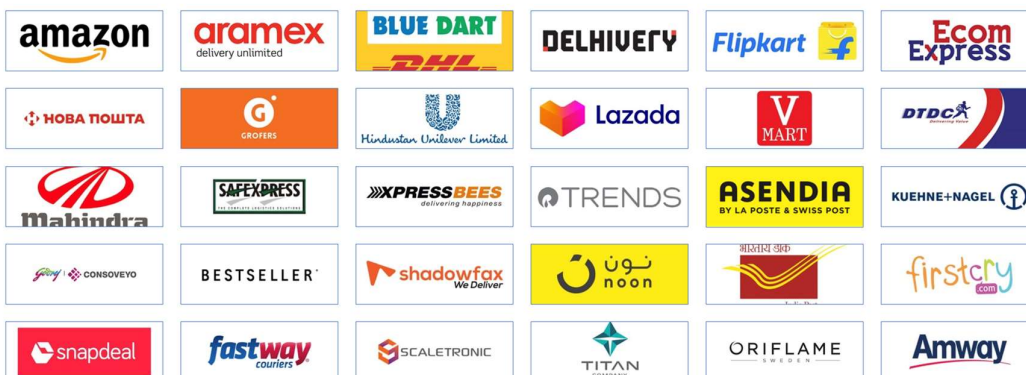


Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements

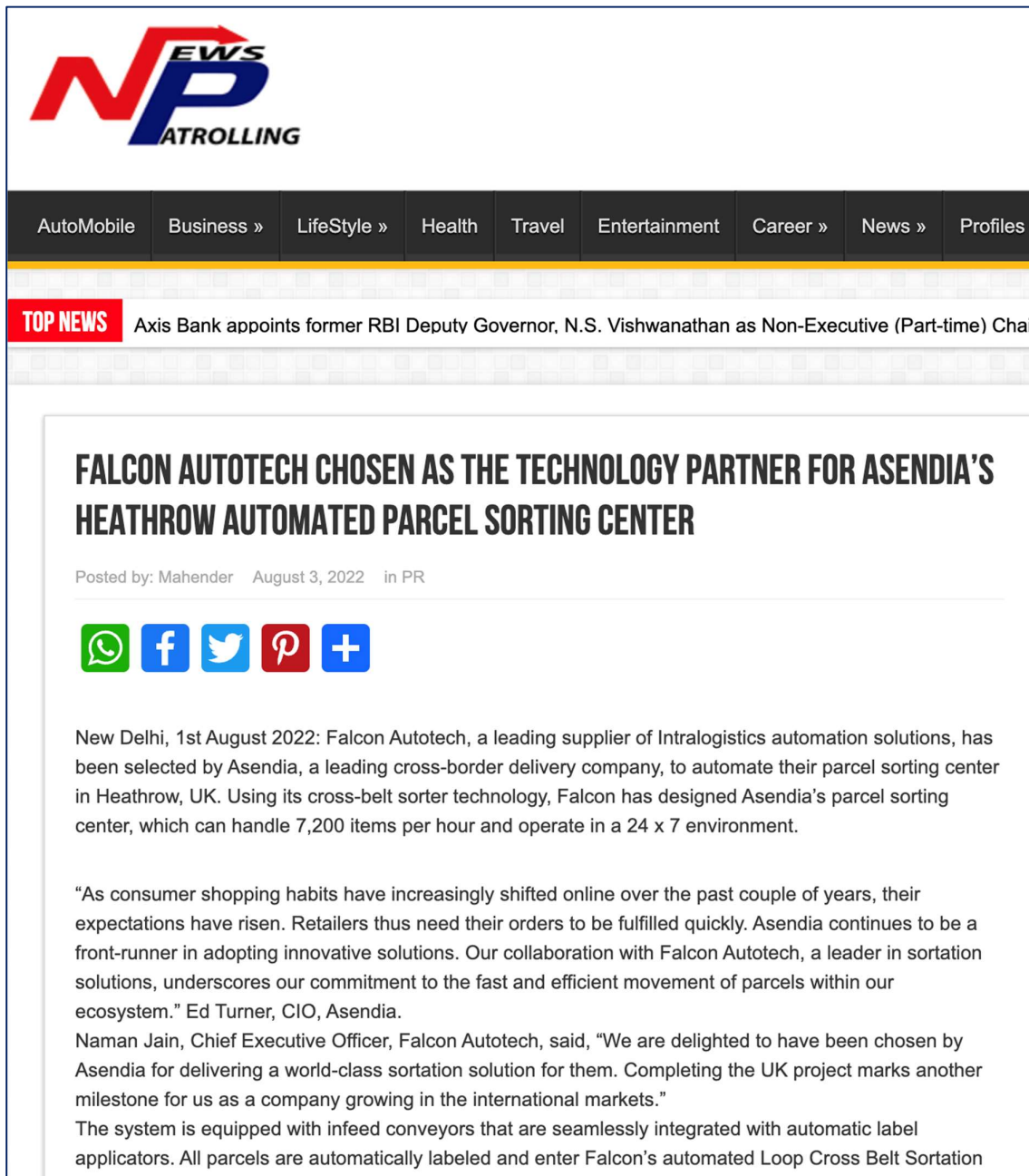


With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.



4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of the **World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Shipments per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Shipment Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focuses on Sortation Solution –

5.1 Project 1- (CEP Client, India)

The System is equipped with two fully automated and interconnected Sub-systems. Sub-System 1 is designed for handling large B2B boxes and E-commerce shipment bags while the Sub-System 2 is designed to handle Small E-commerce packages.

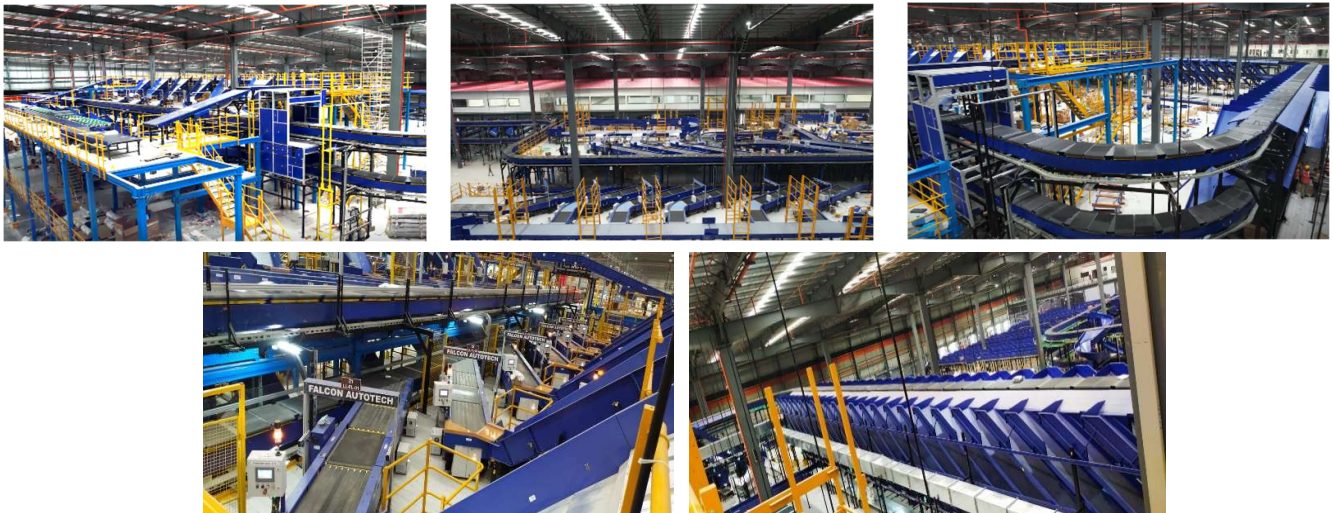
Solution Specifications –

- 48,000 PPH (Double Deck CBS- Shipment Sorter)
- 17,000 PPH (Double Deck CBS- Bag Sorter)
- Building Size: 700,000 Sq. Ft

Key Technology Modules –

- | | |
|---|---|
| <ul style="list-style-type: none"> • 2 Sets of Double Decker CBS Sorters • Mezzanine Structures (one large mezz on to install a double decker Sorter). • Automated Singulators • Fully Automatic Inductions • Semi-Automatic Inductions • Telescopic belt conveyors • PVC Belt Conveyors | <ul style="list-style-type: none"> • Modular Belt Conveyors • Spiral Chutes with Braking rollers • 5-Sided Scanning Tunnels • High speed weighing conveyors • Direct Bagging Chutes • Put to Light Chutes • Volume Distribution systems • High Availability Server Systems • WCS |
|---|---|

Site Pictures -



5.2 Project 2- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

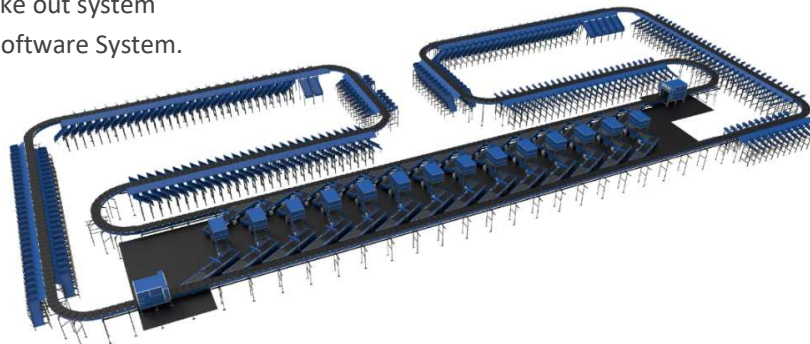
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



5.3 Project 3- (Ecommerce Client, India)

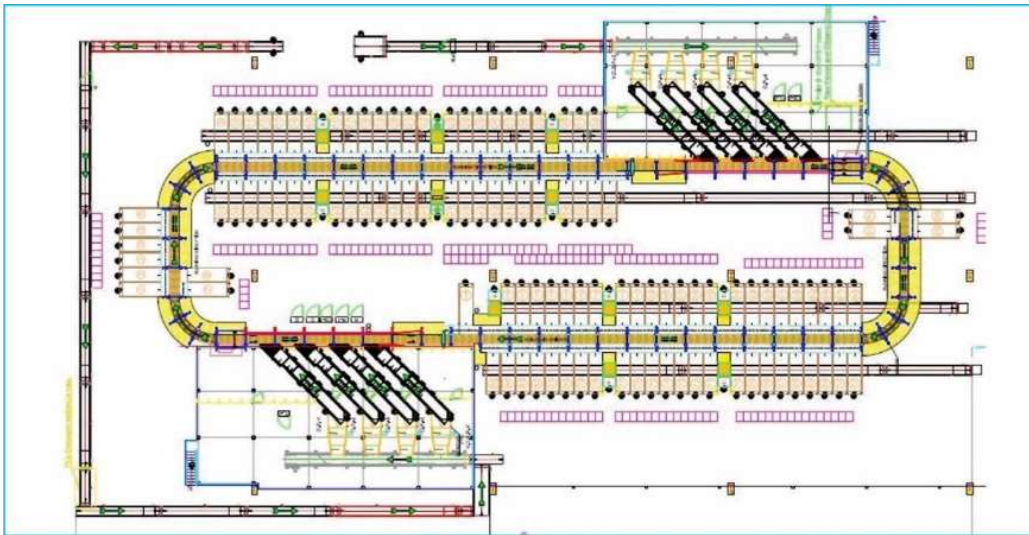
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



5.4 Project 4- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 shipments per hour. The system is equipped with three infeed conveyors integrated with an automatic label applicator before shipments enter the sortation system. The shipments are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.5 Project 5- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 shipments per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the shipments at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight shipments.

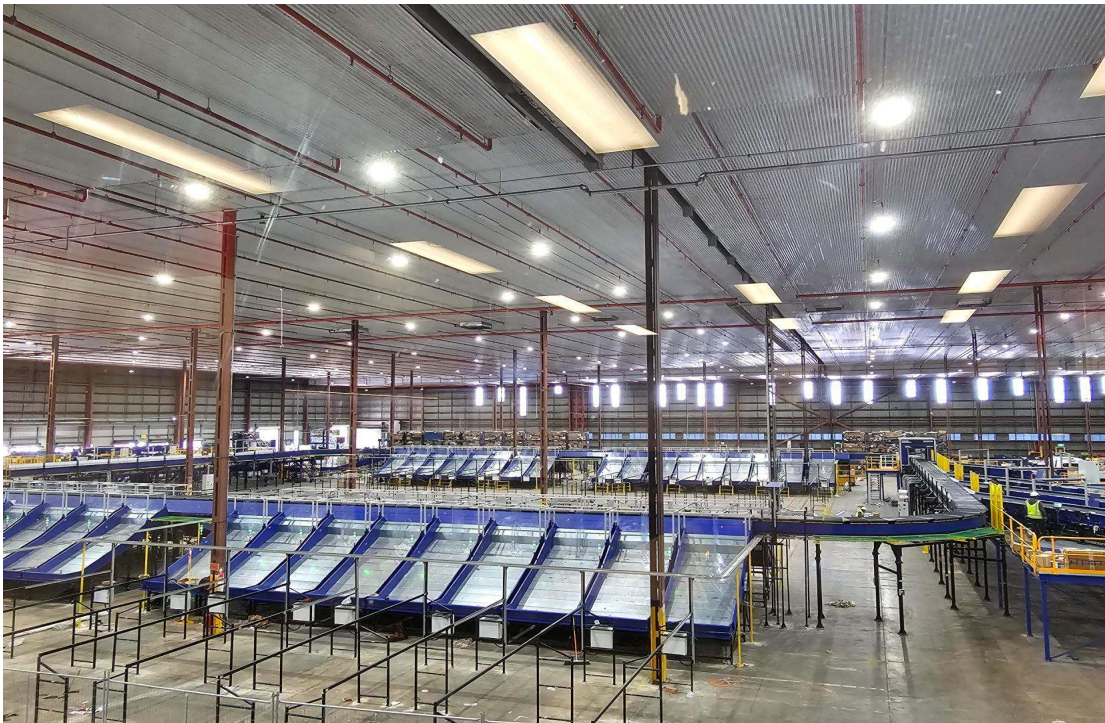
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize shipment.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



6. About Falcon

THE COMPANY

ABOUT US

Falcon Autotech is one of the leading Intra-logistics automation solution providers with a sharp focus on Parcel, Piece and Case Handling applications. We have a broad portfolio of Sortation Systems for Retail, E-Commerce and CEP (Courier, Express and Postal) industries and have supplied our systems across various countries globally.

OUR VISION

To create a global impact on the society by facilitating world's leading organizations run their supply chains efficiently by leveraging technology enabled products and solutions and, in the process, establish Falcon Autotech as one of the world's most respected and trusted brands in Intra-logistics warehouse automation space.

OUR JOURNEY

The team started out in 2004 solving special purpose automation problems for clients and later pivoted to building a standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems. Over the last few years, the organization has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

OUR CORE COMPETENCIES

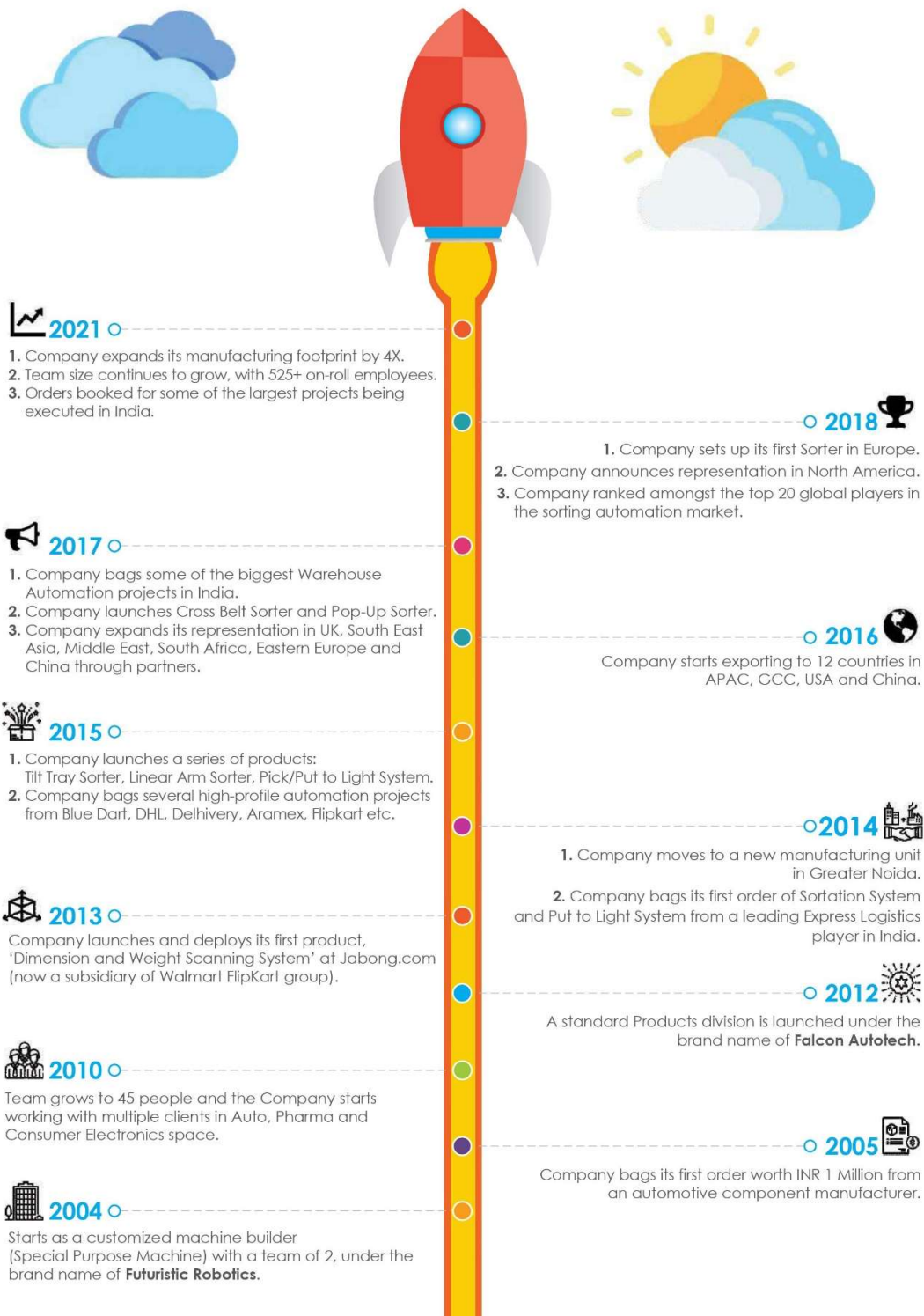
Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more. At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

- ❖ Supply Chain Solution Designing and Consultation
- ❖ Hardware Design
- ❖ Electrical Design
- ❖ Electronics Design
- ❖ Software Development
- ❖ Manufacturing

Giving us an unprecedented ability to harness our true potential generated by culmination of Supply Chain Know-How, quality Hardware and flexible Software under one roof.



COMPANY HISTORY



KEY STATISTICS

**04**Product Lines

**525+**On-Roll Employees

**290+**Engineering Team

**400000+ Sq Ft**Manufacturing Space

**20+**Tier A Clients

**15+**Countries with Live Installations

**1800+**Total Installed Systems Globally

**35**Representations in 4 Continents and 35 Countries

**1Million+**Sorts Per Hour of combined Sorting Capacity
installed Globally

OUR PRODUCT PORTFOLIO

01

SORTATION SOLUTIONS



Cross Belt Sorter



Tilt Tray Sorter



Arm Sorter



Swivel Wheel Sorter



Sweep Sorter



Pusher Sorter



Pop Up Sorter

02

PICK/PUT TO LIGHT SYSTEMS



Packet Sortation



Multi-Line Order Consolidation



Order Picking



Store Order Fulfillment



Spare Parts Distribution



Kitting



Multi Order Pick Cart

03

DIMENSION & WEIGHT SCANNING SYSTEMS

STATIC DIMENSION & WEIGHT SCANNING SYSTEMS



Cubizon-R



Cubizon-R THRU



Cubizon-R Cross



Cubizon-R ECO

DYNAMIC CONVEYOR BASED SYSTEMS

OUR 8 MODELS



OUR PRODUCT PORTFOLIO

04

CONVEYOR AUTOMATION SOLUTIONS

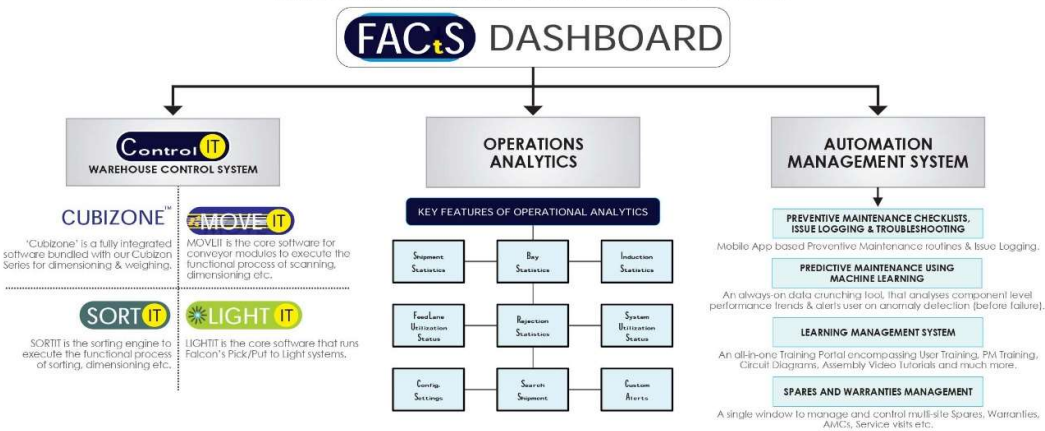
BELT CONVEYORS	ROLLER CONVEYORS	SPECIAL APPLICATION SOLUTIONS
❖ PVC BELT ◆ Straight Conveyor ◆ Spacing Conveyor ◆ Accumulation Conveyor ◆ Strip Belt Merge- 30/45/60 ◆ Solid Belt Merge- 45 ◆ Incline Belt Conveyor ❖ MODULAR BELT ◆ Straight Conveyor ◆ 90 Degree Belt Curve ◆ 180 Degree Belt Curve	◆ Straight Conveyor ◆ Gravity Roller Conveyor ◆ MDR Accumulation ◆ Angled Roller Conveyor ◆ Curved Conveyor ◆ Spur Merge Conveyor ◆ Swivel Wheel Diverter ◆ Pop Up Belt Diverter ◆ Pusher Unit	◆ Aligning Conveyor ◆ One to Two Split Conveyor ◆ Two to One Merge Module ◆ Bulk Distribution-ARB ◆ Bulk Distribution-Telescopic ◆ Bulk Distribution-Electric Arm ◆ Telescopic Belt Conveyor



05

SOFTWARE SUITE

FALCON'S AUTOMATION CONTROL SUITE



ADD-ON MODULES



Real Time Video Coding

No-Read Barcodes have always been a challenge in Sorter Applications globally. With Falcon's RTVC feature, we can now easily decode such challenging parcels in real time while sitting remotely.



Deep Learning Image Classification

DUC is an artificial intelligence function that imitates the working of the human brain. It can identify objects such as packets on the carrier and classify them in different categories.



Cloud Sort App/Virtual PTL

It's an android based app used to perform secondary sorting in light assisted put-away operations. It can work as a stand-alone setup or can be integrated with any sorter.

INDUSTRY SOLUTIONS

As a leading player in the intra-logistics automation space, Falcon Autotech continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



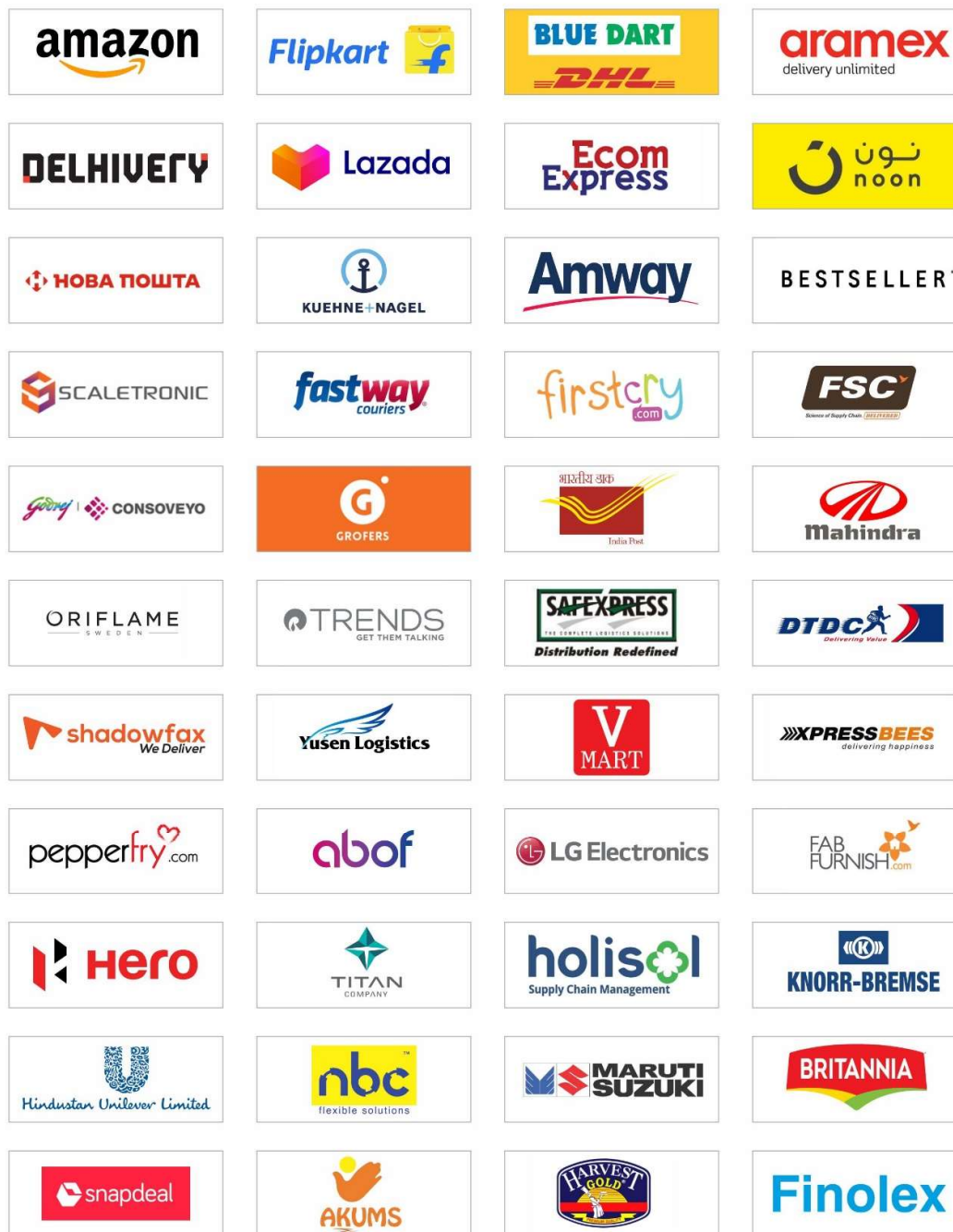
SPARE PARTS DISTRIBUTION



GROCERY DISTRIBUTION

SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.



7. Handled Shipment Spectrum

As per the shipment spectrum data provided, Falcon has studied and analysed the shipment spectrum in detailed.

Falcon purposes to use its “Linear Arm Sorter” to provide the maximum benefits to Glaukous in-terms of handling shipment sizes and weight.

7.1 Shipment size loadable on the sorter

Falcon's Cross belt sorter has a capability to handle the below mentioned shipment sizes and weight.

Specification	Unit	Value
Max Length	mm	508
Max Width	mm	508
Max Height	mm	100
Max Weight	Kg	8
Min length	mm	100
Min Width	mm	100
Min Height	mm	20
Min Weight	gm	150

- **Note:** 90 Percentage shipments are of size 600 x 450 x 450 mm, and 10 percentage are of size 900 x 500 x 500mm

7.2 Shipments to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of shipment cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Shipments shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).

9. The shipments must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

7.3 Shipment not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile shipments with contents not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

7.4 Pictures of Sortable Shipments

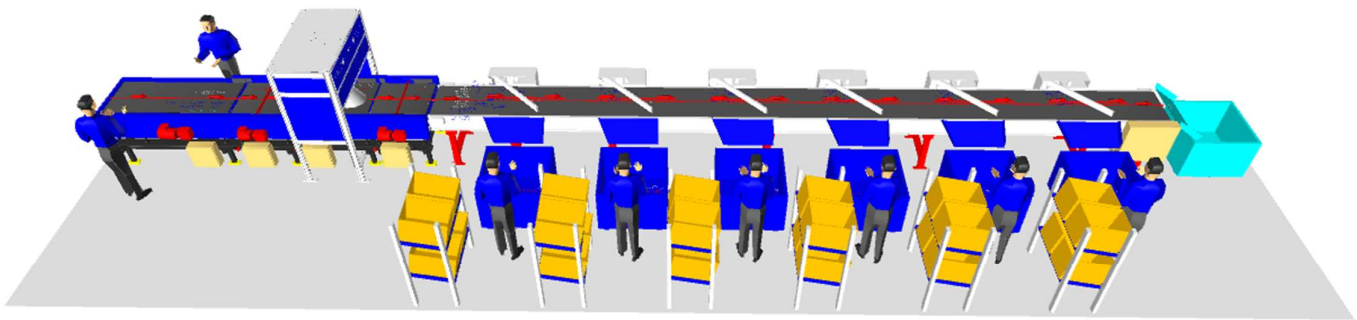
	PARCEL TYPE	DESCRIPTION
	Strongly Strapped	A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (30 mm) may create problems and are not recommended to be processed.
	Other Plastic Covered Item	A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item	A cardboard parcel coated in a packing paper.
	Cloth Covered Item	Often import parcels that are of irregular shape.
	Wine Box	A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box	A cardboard parcel is often the norm.
	Totes	A large plastic, lidded box.
	Flats	A parcel that often contains documents.
	Poly Bags	It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

8. Proposed System Description

8.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the Linear Arm Sorter for sorting shipment as per Glaukous logic.

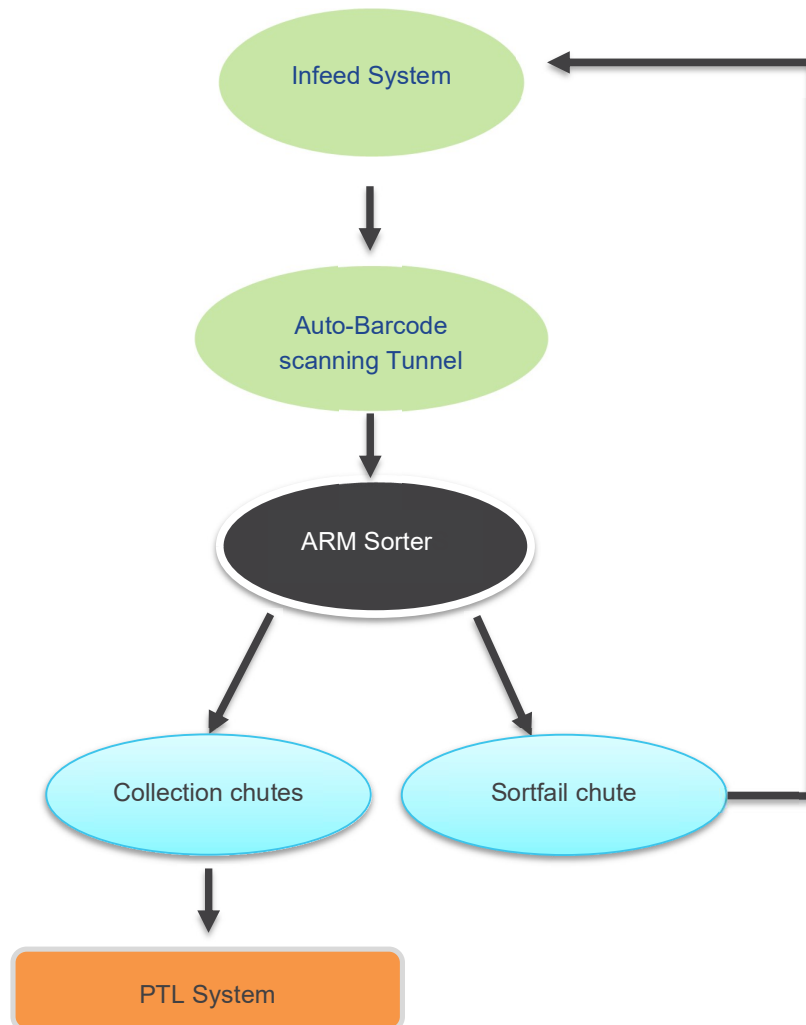
8.2 Summary of the System



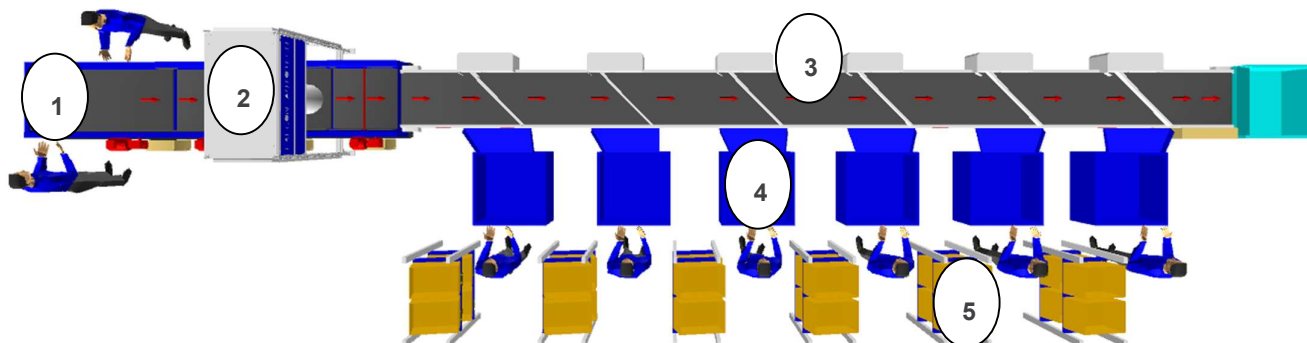
Process flow of the Loop CBS System-

1. **Infeed System:** - Shipments are manually placed in singulated manner onto the belt conveyors, which further travels via set of buffer & spacing conveyor and then reaches Scanning Tunnel.
2. **Arm Sorter:** - Once the shipments reaches scanning tunnel, barcode data is captured & Shipment travels further to Arm sorter conveyor where the Arm diverter efficiently sorts the shipments into their respective output chutes by utilizing the data provided by Glaukous' sorting logic.
3. **PTL System:** - As the shipment are collected in output bin after primary sorting, An operator manually picks the shipment, scan it via Hand held barcode scanner & places it into one of the PTL bin with the help of light indication on PTL modules.
4. **Output Chutes:** - The shipments are discharged into two types of chutes.
 - a. **Collection Chute** – In the Arm sorter there are 6 collection chute (Bin type) to accumulate the shipments dropped via Arm diverters.
 - b. **PTL Bins** – Four Bins are linked to each collection chute in order to carry shipment undergoing secondary sorting.
 - c. **Sortfail Chute**- One sortfail chute is present to handle sortfail Shipments.

9. Concept Description



10. Layout View



Legend

- 1 – Infeed Line
- 2 – Auto Barcode scanning tunnel
- 3 – Arm sorter
- 4 – Collection chute (Bin Type)
- 5 – PTL System

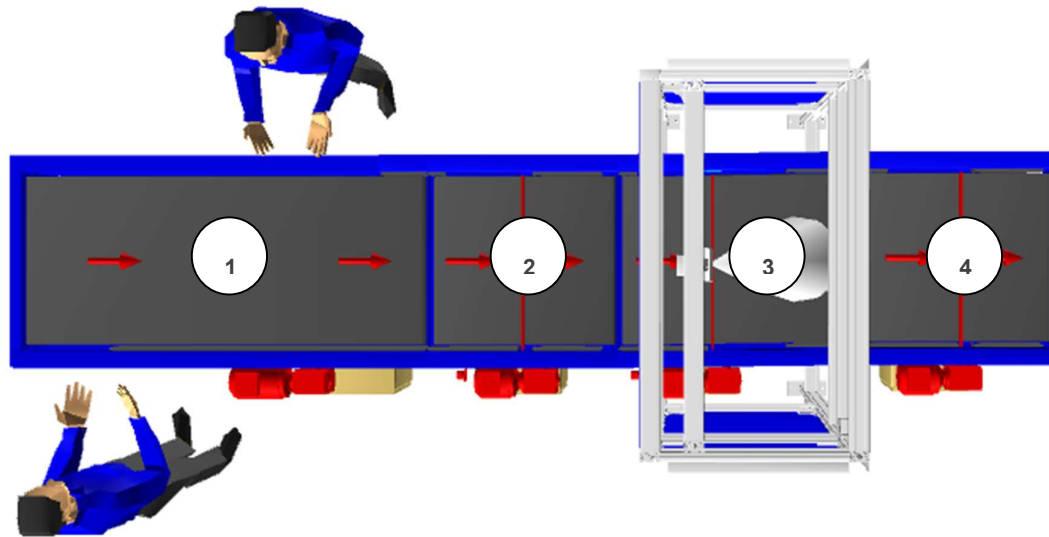
11. Proposed System Capacity Calculations

The following table shows the throughput calculation for the sortation system designed based on Glaukous requirements.

SPECIFICATION	VALUE
Sorter Type	ARM Sorter
Pitch	1,700 mm
Effective Designed Throughput of Arm Sorter	2,000 Shipments per hour
Per Feedline Capacity	2,000 Shipments per hour
No of Feedlines	01 Nos.
Total Feedline designed capacity (B)	2,000 Shipments per hour

12. System description

12.1 Infeed System



Legend

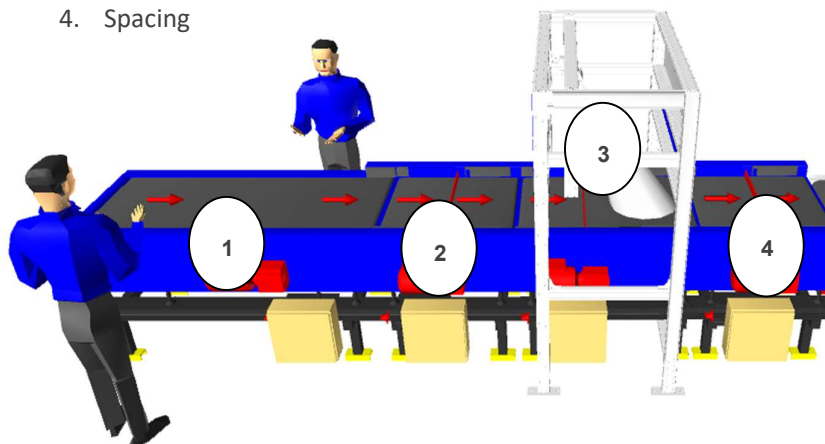
- 1- Loading Conveyor
- 2- Buffer Conveyor
- 3- Scanning Tunnel
- 4- Spacing Conveyor

The Arm sorter facilitates One Infeed lines to transport the shipments from initial point to Highway conveyor for sorting.

12.1.1 Infeed System – (Description)

Infeed conveyor system consists of below components:

1. PVC Belt Conveyor
2. Buffer Conveyor
3. Auto-Scanning Tunnel
4. Spacing



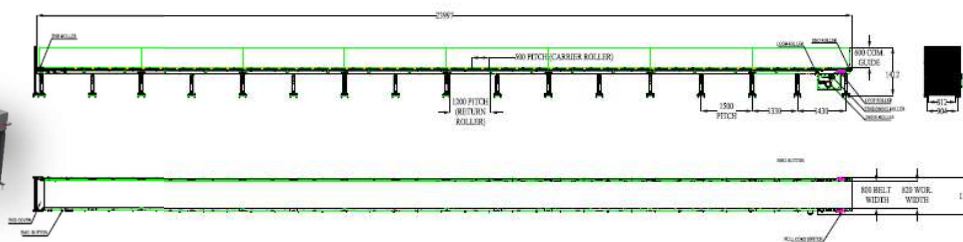
12.1.1.1 Straight Powered Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

Some Silent Features of Falcon's Belt conveyor:

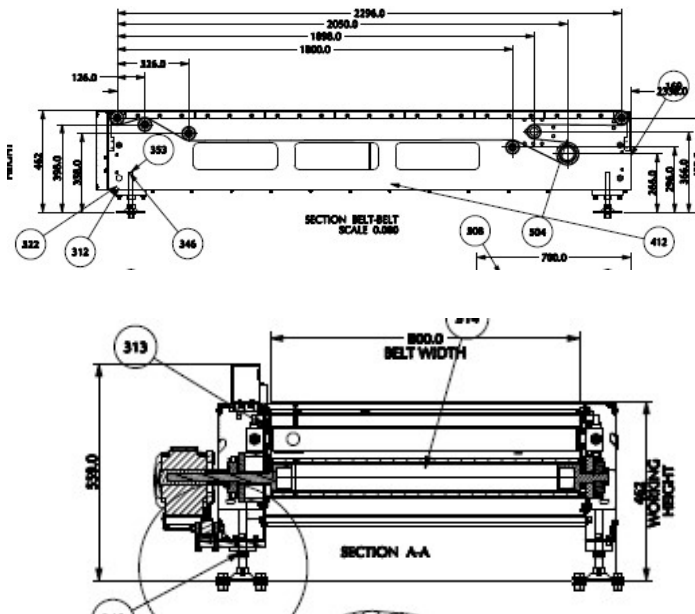
- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



12.1.1.2 Buffer Conveyors

A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

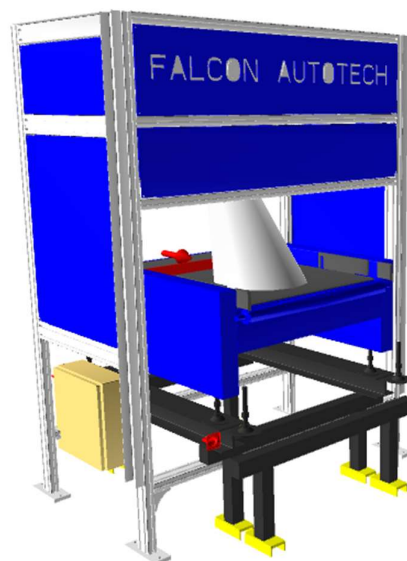
This Conveyors required to maintain the Throughput of Line.



12.1.1.3 Auto Scanning Tunnel

Fully Automatic top Side scanner is provided for best data read rate, enhancing overall Sorter efficiency.

Make - Cognex



12.1.1.4 Spacing Conveyor

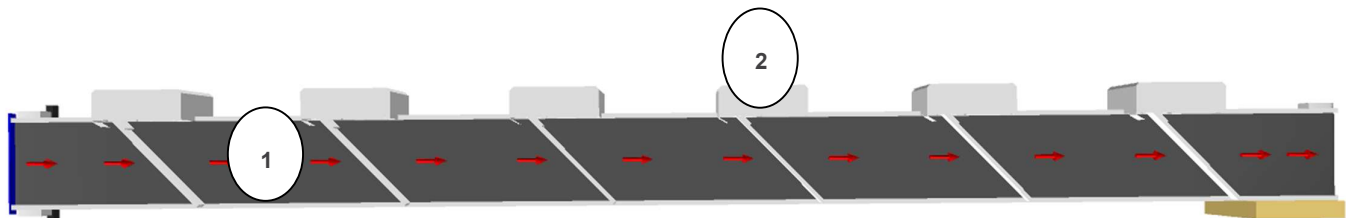
It is a Dual set of high-speed conveyors, using to create appropriate space between parcels/boxes.

Some Salient Features of Falcon's spacing conveyor:

- Low Noise
- Maximum uptime.
- High safety standards.
- Fastest ROI
- Comes with MS frame.



12.2 Main Arm Sorter



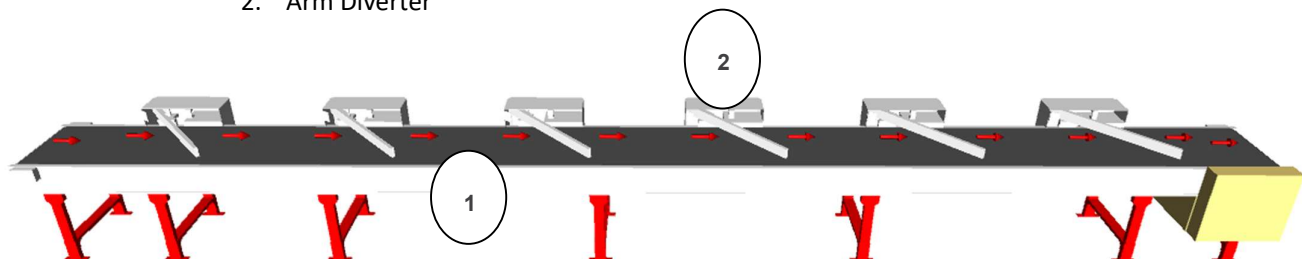
Legend

- 1- Modular conveyor
- 2- Arm Diverter

12.2.1 Arm Sorter – (Description)

Infeed conveyor system consists of below components:

1. Modular Conveyor
2. Arm Diverter

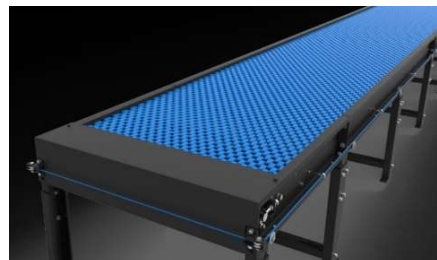


12.2.1.1 Modular Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Modular Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



12.2.1.2 Arm Diverter

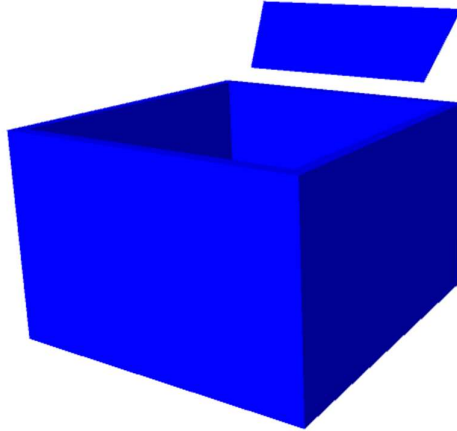
Arm diverter System of Falcon Autotech is an extremely cost-effective Sortation system. Arm diverter's simple and modular design gives it advantage of low cost, space saving, and modularity.



Specification	ARM Regular
Actuation method	Electric induction motor
Belt type	Flat top modular
Belt width (mm)	~700
Arm length (mm)	1000
Max. product weight (kg)	10

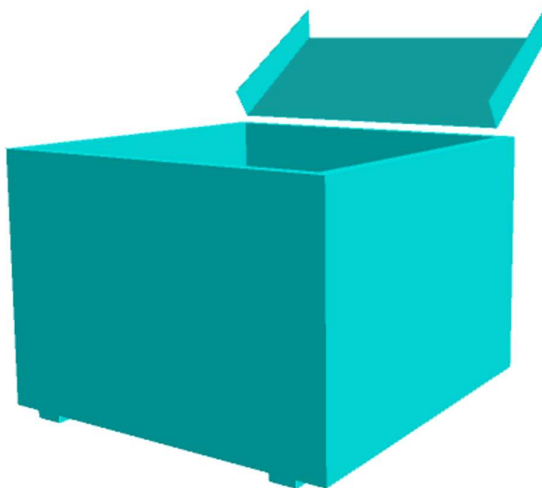
12.3 Output Chute

12.3.1 Collection Chute (Primary)



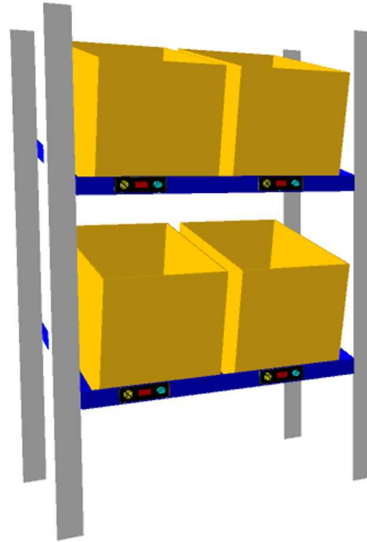
Total Six Output chutes are provided in the system, to accommodate sorted parcels from Arm Sorter

12.3.2 Sortfail Chute

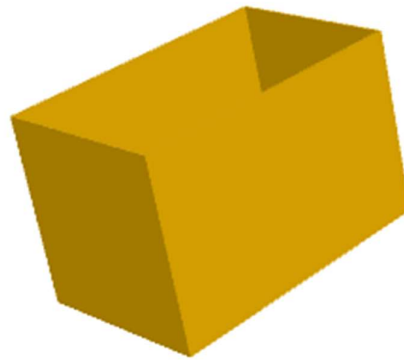
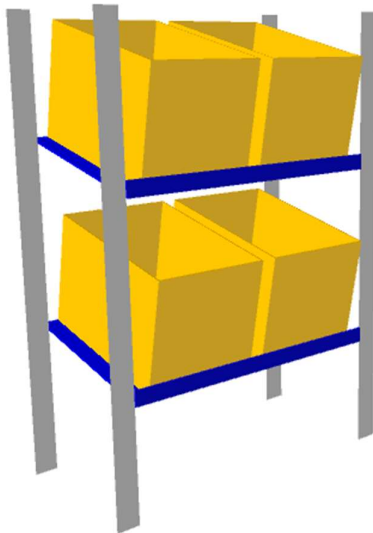


One Sortfail chutes is provided in the system, to accommodate sortfail parcels those failed to get sorted in either chutes.

12.4 PTL System



12.4.1 PTL Bins



12.4.2 Put to Light Modules

PTL system is falcons in house developed modular light hardware modules and a connected software platform to ensure a flexible, robust, seamless, and easy to setup PTL solution which is fully integrated with Client's WMS/ERP Systems.

The Software platform is powered by a deep artificial intelligence engine that continually mines the data generated by operations and keep on altering the PTL configuration to increase the system productivity and give you the extra competitive edge.

The System involves a collection of physical sorting locations representing either a customer order/ address location or any other logical sorting group. Each physical location is mapped to a light module which glows on the event of scan of a product for that sorting location.

It is a paperless technology provide high throughput with single operator with high accuracy



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Dimensions	Mm	Around 165 X 50
LED Display	Type	3 Digit Alpha numeric Rotating Display
LED Size	Mm	~ 25 X 20
LED Indicator	yes/no	YES
Acknowledgment Button	yes/no	YES
Power Requirements	Type	3 Phase UPS Power
Communication	Type	TCP/IP and Serial
Operating temperature	Deg.	45 Degrees

12.4.3 Hand Held Scanner (2D)



13. Proposed System Technical Details/BOM

13.1 Mechanical equipment

Pos.	Qty.	Description	Value
1	1	Infeed System - Loading Conveyor - Buffer Conveyor - Scanning Tunnel - Scanning Conveyor - Spacing Conveyor	01 Nos 01 Nos 01 Nos 01 Nos 01 Nos
2	1	ARM Sorter - ARM Diverters - Modular Conveyor - Sorter Height - Sorter Length - Sorter Speed	06 Nos ~11 metres (1 Modules) 1000mm (from ground level) ~17 m ~1 m/s
3	1	Sorter Outputs - Collection chute - Sort fail chute - Transfer Plates	06 Nos 01 Nos 07 Nos
4	1	PTL System - Hand Held scanner (2D) - PTL Modules - PTL Bins	06 Nos 24 Nos 24 Nos

13.2 Electrical Equipment

Electricals			
1	1	Electrical Cabling and Wiring Main Power Distribution panel Main Control Panel Network Switches Field Cabling	Included

13.3 Control System

Components			
1	1	Controls and IT Infrastructure Controls Weigh Scale Control, Barcode Scanner Controls, Control IT Software Package	Included

14. Electrical System

Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.

PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets **Engines**

Three-phase alternating current motors (Induction) will be used through a frequency converter.

The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.

All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

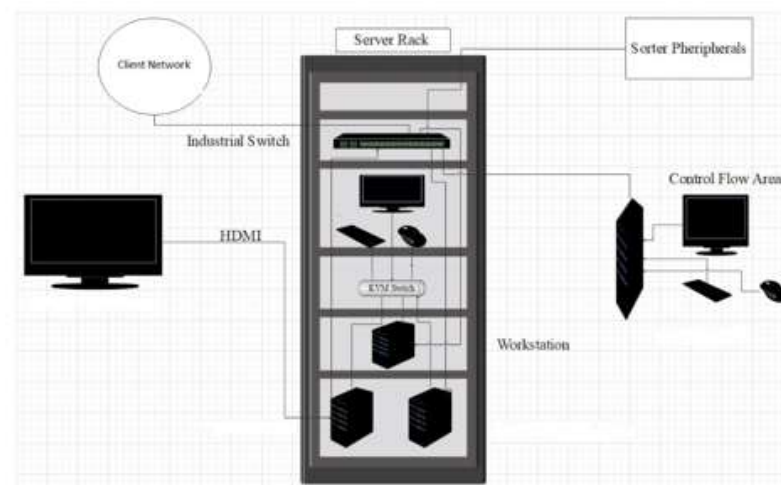
Control command

The proposed solution is based on OMRONS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be Ethercat.

Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the Ethercat field bus

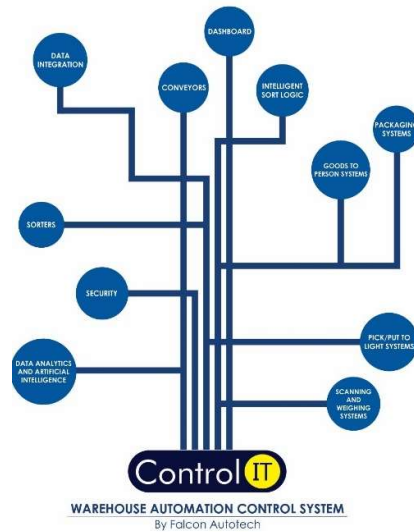
15. IT Architecture (Ref.)



15.1 HAA Server (In Glaukous Scope)

S.no	Description	Qty
20Core Config with 128 GB RAM in T440 and 64 GB RAM in T40		
1	Synology_storage_DS723+ 2.6 GHz AMD Ryzen R1600 Dual-Core, 2 x Gigabit Ethernet Ports,	2
	2GB ECC DDR4 RAM, 2 x 3.5/2.5" Bays 2 x M.2 2280 Slots E10G22-T1-Mini 10GbE RJ-45 network upgrade module for compact Synology servers	
2	Dell Tower Model T440 -PowerEdge T440 Xeon Gold 6148 2.4GHz/20C/27.5MB/150 W 16 DIMMS 4 x32GB RDIMM Up to 8 3.5"" Hot Plug Hard Drives Tower Configuration 2 x 1.2TB 10K RPM SAS 12Gbps 512n 2.5in	2
	Hot-plug Hard Drive PERC H750 Adapter Full Height 8GB Cache Dual Hot-plug PS495W iDRAC9Enterprise 3YR ProSupport Next Business Day Onsite/Dual Port Lan	
3	Dell PowerEdge T40 Intel Xeon E-2224G Processor 3.5GHz 8M Cache,4C/4T,Turbo,71W,TPM, 4*16 GB RAM (4 DIMM), 2x1TB SATA 3.5" 7.2k rpm HDD (3 Bay), 1Gbe LOM ,DVD Writer, Onboard RAID 01, Inbuilt PSU Standard (Max 1), 3	1
	Year Onsite NBD,480 GB SSD,1 Gbe Dual Lan Card Extra	
4	Windows Server 2019	3
5	Monitor	1
6	KVM Switch	1
7	Mouse	1
8	Keyboard	1
9	Lan Cable	10
10	Power Cable	8
11	VGA Cable	1
12	42 AC Server Rack	1
13	Netgear 24 Port Giga Switch	2
14	2 TB SSD Micron	4
15	DP to VGA Convertor Cadyce Brand	1

16.Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

17. Key Components Make

Items	Make
Belts	Forbo/ Derko
Bar Code Scanner	Cognex
Sensors	Sick/Leuze/ P&F
PLC	Omron
Control Panels	Rittal/ BCH
VFDs	Omrons/Lenze/ AB/ Omron
Cables	LAPP/ Equivalent
Switch Gear	Schenider/Equivalent
Bearings	NTN/SKF/Equivalent
Power Transmission Systems	Vahle
HMIs	Omrons

18. Program Organisation

18.1 Program/ Project Schedule

18 -20 weeks Ex-works greater Noida.

**Above plan is tentative, detailed plan will be shared post order confirmation*

18.2 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with Glaukous
3. Weekly/Fortnightly Meeting Glaukous to share the Project Status.
4. Scheduling of the resource management.
5. Management of risks and opportunities.
6. Management of the requirements.
7. Management of the list of anomalies or reservations.

19. Commercial Terms

19.1 Price Sheet

S. No	Component	Set	Price (INR)
1	Arm Sorter	1	₹ 42,86,087
2	Conveyor Automation	1	₹ 2,52,000
3	Destination	1	₹ 1,07,408
4	PTL System	1	₹ 4,39,680
5	Toolkit	1	₹ 36,000
6	Software Packages	1	₹ 2,00,000
7	Installation & Commissioning	1	₹ 2,04,847
8	Packaging & Forwarding	1	₹ 1,02,424
9	Project Management and Engineering Charges	1	₹ 84,427
Grand Total (INR)			₹ 57,12,872

The Total Price is to be understood.

- Ex Works – Greater Noida, India
- Freight charges shall be extra at actual.
- Taxes extra at actual.
- Including Packing and Forwarding
- Including Installation and Commissioning
- Price is valid for 30 days from the date of proposal

*Material Unloading & Laydown area is in customer scope

19.2 Payment Terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 70% Advance along with PO.

Stage 2: 30% along with 100% Taxes prior to dispatch.

Stage 3: 20% Within 30 days of commissioning.

20. Warranty Period

Falcons offered System comes with a standard warranty of 1 year and post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Faulty articles:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

21.Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Low Spec Server PC**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Server PC is not included in this proposal
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.